

# Homework 3

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##Set Up

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
```

```
# creating object for kelp data
kelp <- read_csv("temp-kelp.csv")
# creating object for personal data
commute <- read_csv("envs193ds-personal_data - Sheet1.csv")
```

## Problem 1

a.

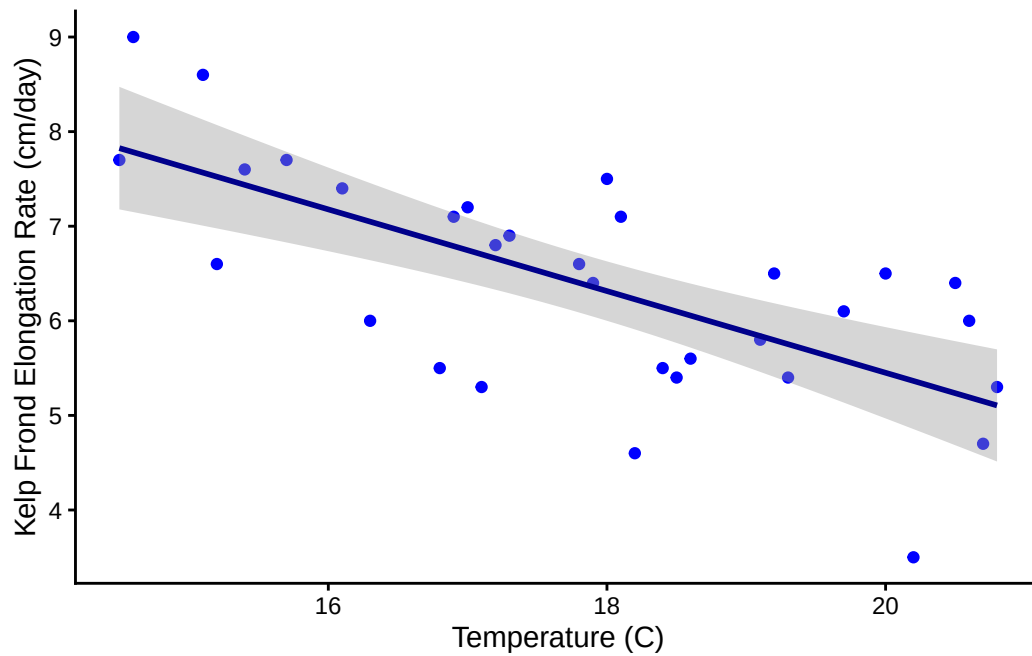
The two tests that we should run to determine the relationship between temp. and giant kelp elongation rate would be a Pearson correlation test and a Spearman rank test. The tests both measure the strength of the relationship between the variables but the Pearson test relies on a normal and linear distribution and the Spearman test does not need normal distribution.

b.

```
# base layer ggplot
ggplot(kelp, mapping = aes(x = temp_c, y = kelp_elong)) +
  # first layer: scatter plot
```

```
geom_point(color = "blue") +
# second layer: line of best fit + ribbon
geom_smooth(method = "lm", se = TRUE, color = "darkblue") +
# renaming x and y axis
labs(x = "Temperature (C)", y = "Kelp Frond Elongation Rate (cm/day)") +
# changing the theme of the graph to classic
theme_classic()
```

`geom\_smooth()` using formula = 'y ~ x'



c.

### Checking assumptions

Linear = yes the data does have a linear relationship. You can see this in problem 1b where the linear graph is the right model for the data.

Normality = Yes the data is normally distributed, this is shown through the Shapiro test.

Based on these two assumption checks the correct test to run would be the Pearson correlation test.

```
shapiro.test(kelp$temp_c) # running normality test for temp
```

Shapiro-Wilk normality test

```
data:  kelp$temp_c  
W = 0.96254, p-value = 0.3219
```

```
shapiro.test(kelp$kelp_elong) # running normality test for kelp growth rate
```

Shapiro-Wilk normality test

```
data:  kelp$kelp_elong  
W = 0.98756, p-value = 0.966
```

#### running the test

```
# running the correlation test  
cor.test(kelp$temp_c, # setting first variable  
         kelp$kelp_elong, # setting 2nd variable  
         method = "pearson") # setting the method to be Pearson
```

Pearson's product-moment correlation

```
data:  kelp$temp_c and kelp$kelp_elong  
t = -5.189, df = 30, p-value = 1.366e-05  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
-0.8359665 -0.4460157  
sample estimates:  
cor  
-0.6877489
```

**d.**

In order to test the strength of the relationship between temperature in Celcius and kelp frond elongation rate in cm/yr I used a Pearson test of correlation ( $r(30) = -0.687$ ,  $p = 1.36 \times 10^{-5}$ ,  $\alpha$

= 0.05, CI = (-0.85, -0.45)). This test was ran because the data was normally distributed and showed a linear relationship. This test yielded a value of  $r = -0.69$ , which shows a moderate, negative linear correlation between variables. Because of this value and the p value being less than 0.05 we can safely reject the null hypothesis on the grounds of there being a negative correlation between the variables.

e.

The results seem to show that the increasing temperature of the ocean may cause kelp growth rates to slow down, which would in turn have larger effects on the kelp's overall health. When the kelp was grown in warmer temperatures the kelp grew at a slower rate. The relationship is represented by a value of -0.69 which means that there is a moderately strong, negative correlation between temperature and elongation rate.

f.

```
# setting up the correlation test
cor.test(kelp$temp_c, # setting first variable
         kelp$kelp_elong, # setting 2nd variable
         method = "spearman") # assigning spearman test
```

Spearman's rank correlation rho

```
data: kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

Through running a Spearman rank test ( $\rho = -0.69$ ,  $p = 1.29 \times 10^{-5}$ ,  $\alpha = 0.05$ ) I determined that there is a strong negative relationship between temperature and frond elongation rates. The two tests have both led me to reject the null hypothesis of there being no correlation between variables because they both show a negative relationship between the increasing temperature and decreased growth rates. The Spearman test tested without normality and found there is a negative relationship, and the Pearson test used normality and also found a negative, linear relationship.

## Problem 2.

```
commute_summary <- commute |>
  clean_names() |>
# grouping by categorical variable
group_by(day_of_the_week) |>
# calculating mean and sd
summarize(
  mean = mean(commute_length_min),
  sd = sd(commute_length_min))
```

```
# creating object to clean column names
commute <- commute|>
  # removing spaces and parentheses in column names
  clean_names()
```

```
# base layer: ggplot
ggplot(data = commute,

# assigning x = days of the week, y = commute length
mapping = aes(x = day_of_the_week,
               y = commute_length_min,
               color = day_of_the_week)) +

# first layer: jitter
geom_jitter(height = 0,
            alpha = 0.5,
            width = 0.5) +

# second layer: point range
geom_pointrange(data = commute_summary,

# assigning the values of mean and sd
mapping = aes(x = day_of_the_week,
               y = mean,
               ymax = mean + sd,
               ymin = mean - sd)) +

# labeling axis and title
labs(x = "Day of the week",
      y = "Commute Length in Minutes",
```

```

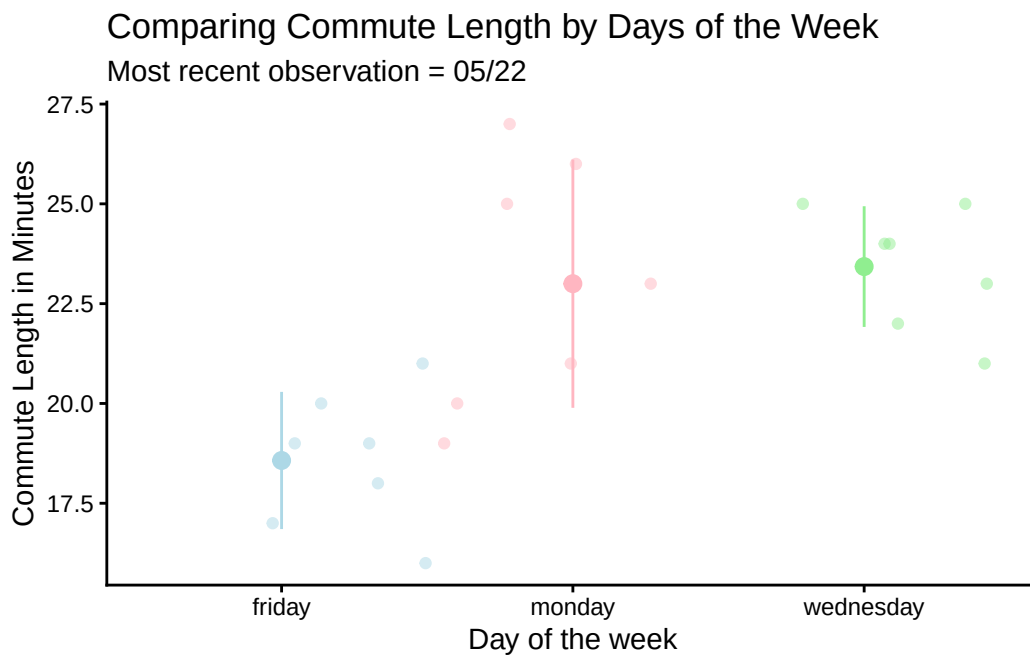
    title = "Comparing Commute Length by Days of the Week",
    subtitle = "Most recent observation = 05/22") +

# changing theme
theme_classic() +

# removing legend
theme(legend.position = "none") +

# changing color
scale_color_manual(values = c("lightblue",
                              "lightpink",
                              "lightgreen"))

```

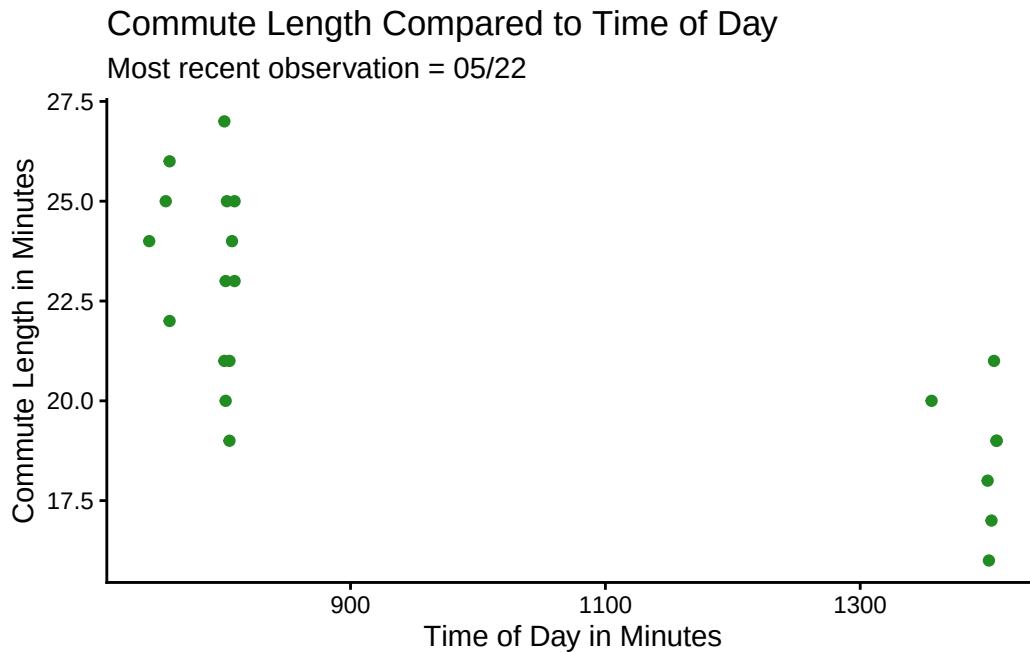


```

# base layer: ggplot
ggplot(data = commute,
# assigning x and y axis
mapping = aes(x = time_of_day_hr_min ,
y = commute_length_min)) +
# adding in the data points and coloring
geom_point(color = "forestgreen") +
# changing axis labels and adding titles

```

```
labs( x = "Time of Day in Minutes",
      y = "Commute Length in Minutes",
      title = "Commute Length Compared to Time of Day",
      subtitle = "Most recent observation = 05/22") +
# retheming
theme_classic()
```



**b.**

Plot 1. Jittered plot showing the mean and standard deviation for commute length depending on the day of the week, Light blue data points represent Friday commutes, pink represents Monday, and green represents Wednesday. The Monday and Wednesday commutes have a higher mean length than the Friday data. Each jittered point represents a specific data point, and the large points and error bars represent the mean and standard deviations for each day. Plot 2. Scatterplot depicting relationship between time of day and commute length in minutes. Each point shows a single data point representing a commute length. The plot shows that as the time increases, the length of the commute decreases, showing a negative relationship between time of day and commute length in minutes.